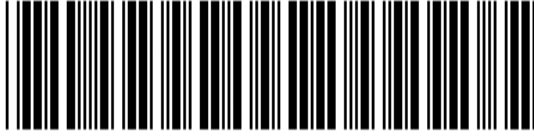
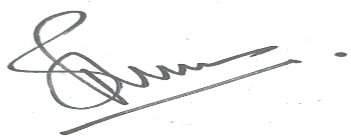


Joint Entrance Examination (Main) – 2025 Session 2
Admit Card-Provisional

Roll Number:	WB10203193	Application Number:	250310436848	 Photograph of candidate
Candidate's Name:	ARGHYAJEET BASAK	Father's / Mother's Name:	PARTHA PRATIM BASAK	
Gender:	Male	Date of Birth:	09-08-2006	
Category:	OBC- NCL (Central List)	State of Eligibility:	WEST BENGAL	
Person with Disability : (PwD/ PwBD)*	No	Scribe required:	--	
	Roll Number Barcode 			Arghyajeet Basak Signature of candidate
Test Details				
Applied For	Paper-1 (B.E./B. Tech.)			
Medium of Question Paper	Date of Examination	Shift	Timing of Test (Indian Standard Time) (IST)	
English	02.04.2025	Second	03.00 P.M. to 06.00 P.M.(IST)	
Reporting Time at Centre	01.00 P.M.(IST)	Entry Gate Closing Time at Centre	02.30 P.M.(IST)	
Test Centre Code, Name and Address	WB1001 TCS Gito Bitan Block DN, Plot No 54 and 55, Sector V , Bidhan Nagar East, Salt Lake, Kolkata, West Bengal, India, 700091	Photo ID uploaded by candidate@	Type	Aadhaar
			Number	**** * 2700
		 DIRECTOR (EXAMS), NTA		

@The candidates are advised to bring photo ID proof as uploaded in online application at the time of registration as given above.

SELF DECLARATION (UNDERTAKING)

I, **ARGHYAJEET BASAK**, resident of **F-35 SCHOOL ROAD SCHOOL ROAD NORTH 24 PARGANAS WEST BENGAL 700110**, do hereby, declare the following:

- I have read the Instructions, Guidelines, Information Bulletin, and Notices related to this examination available on the website <https://jeemain.nta.nic.in/> and www.nta.ac.in
- I have read the detailed "IMPORTANT INSTRUCTIONS for CANDIDATES" as given on Page 2, and I undertake to abide by the same.
- I have also read Important Advisory as given on Page 3.
- The candidates were advised to register through Aadhaar/Digi Locker/PAN Card/Passport. Those candidates who did not register through Aadhaar (or chose authentication through non-Aadhaar options), need to report early on the day of the examination (at least 1 hour earlier from the reporting time) to get their Biometrics recorded at the Exam Centre.

Candidate's Photo (Same as uploaded on the Application Form to be affixed before reaching the Centre)	Candidate's left-hand thumb Impression (To be affixed before reaching the Centre)	Candidate's Signature (To be signed, on the Day of Examination in presence of the Invigilator only)
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The above undertaking has to be filled up in advance before reaching the Centre, except for the candidate's signature which has to be affixed in the presence of the Invigilator.

**Joint Entrance Examination (Main) – 2025 Session 2
Admit Card-Provisional**

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. The candidate must reach the Centre at the time as indicated against Reporting/Entry time at the Centre, in the Admit Card.
2. **No Candidate should adopt any unfair means or indulge in any unfair examination practices. If any candidate is found indulging in any unfair means/unfair examination practices at any stage of examination, his/her candidature will be cancelled, and legal action will be taken including debarring in all future examinations conducted by NTA.**
3. No candidate shall be permitted to enter after the Gate Closing Time.
4. No candidate shall be permitted to leave the Examination Room/ Hall before the end of the examination.
5. Upon completion of the examination, please wait for instructions from the Invigilator and do not get up from your seat until advised. The candidates will be permitted to move out, one at a time only.
6. All candidates are required to download and read carefully the Instructions given with the Admit Card and strictly adhere to them.
7. This Admit Card consists of four Pages - **Page 1** contains the Centre details and Self Declaration (Undertaking), **Page 2** has "Important instruction for candidates", and **Page 3** has "Important Advisory", **Page 4** is a Declaration from Non-Aadhaar Candidates.
8. The Admit Card is provisional, subject to satisfying the eligibility conditions as given in the Information Bulletin.
9. Candidates are advised to verify the location of the test venue a day in advance so that they do not face any problems on the day of the test.
10. If religion/customs require you to wear specific attire, please visit the Centre early for thorough checking and mandatory frisking.
11. No Candidate would be allowed to enter the Examination Centre, without Admit Card and undertaking, Valid ID Proof, and proper frisking.
12. All the Examination Centres are under surveillance of CCTV and equipped with Jammers.
13. **Candidates will be permitted to carry only the following items with them into the examination venue:**
 - a. Personal transparent water bottle,
 - b. A simple transparent Ball Point Pen
 - c. Admit Card downloaded from NTA website (a clear printout on A4 size paper) duly filled in.
 - d. Additional 01 passport-size photograph for pasting on the Attendance Sheet
 - e. Original valid ID proof as uploaded in the online application form.
14. Before reaching the Centre, candidates must enter the required details in the Self Declaration (Undertaking) in legible handwriting, paste the Photograph, and put a thumb impression at the appropriate place on the Admit Card. They should ensure that their Left-Hand Thumb Impression is clear and not smudged.
15. The candidates are advised to bring photo ID proof as uploaded in online application at the time of registration.
16. **PwD/PwBD candidates must bring a PwD/PwBD certificate issued by the Competent Authority if claiming relaxation under the PwD category. The Scribe (provided by the NTA/ brought by the candidate) must also carry his/ her own Self Declaration (Undertaking) regarding educational qualification, passport-size photograph, and valid government identity.**
17. **Please note that the examination will be held according to Indian Standard Time (IST) for all candidates, including those taking it at centers outside India.**
18. **For the Drawing Test—Part III of B.Arch., the candidate must bring their own Geometry Box Set, Pencils, Erasers, and Colour Pencils or Crayons. The Candidates are not allowed to use watercolours on Drawing Sheet.**
19. Candidates are **NOT allowed to carry any other personal belongings, including electronic devices, or mobile phones** to the Examination Centre. Examination Officials will not be responsible for the safekeeping of personal belongings and there will be no facility.
20. **Shoes/footwear with thick soles and garments with large buttons are NOT permitted.**
21. **If a candidate goes for bio-break /toilet, he/she should undergo the mandatory frisking and biometrics again.**
22. **Apart from the biometric attendance and frisking at entry, candidates will be frisked and biometric attendance will be taken again on entry from bio- break/toilets break.**
23. Blank paper sheets for rough work will be provided in the examination Hall/Room. Candidates must write their name and Roll Number at the top of each of the sheet(s) and must drop the sheet/s in the designated drop box without fail, before leaving the Examination Hall/Room. Failures to do so may result in a non- evaluation of your answers.
24. Duly filled-in Admit Card must be dropped in the designated DropBox while leaving the Examination Hall/Room. Failures to do so may result in a non-evaluation of your answers.
25. The candidate should ensure that the Question Paper available on the computer screen is as per his/her opted subject/medium indicated in the Admit Card. In case the subject/medium of the Question Paper is other than his/her opted subject/medium, the same may be brought to the notice of the Invigilator concerned.
26. If at any stage, it is found that the candidate has submitted multiple applications, his/her candidature will be cancelled, and legal action will be taken including debarring in all future examinations conducted by NTA.
27. Candidates are advised to check updates on NTA's website, i.e., www.nta.ac.in / <https://jeemain.nta.nic.in/> regularly. They should also check their mailbox on the registered-mail address and SMS in their registered Mobile No. for the latest updates and information regarding the exam.
28. For any clarification/assistance, you can write to NTA at jeemain@nta.nic.in or call at Helpline number **011 – 40759000, 69227700**.

**Joint Entrance Examination (Main) – 2025 Session 2
Admit Card-Provisional**

IMPORTANT ADVISORY

Dear Candidates,

"We wish you all the best for your examination"

The main purpose of this **Advisory** is to make you aware that the National Testing Agency aims to conduct fair and transparent examinations so that deserving candidates only get the right admissions or jobs. Therefore, it is advisable to refrain from using any unfair means or practices to gain any unfair advantage over other candidates.

1. Kindly note that in many cases;

- i. Candidature has been cancelled and candidates have been debarred for 3 years from appearing in the exams.
- ii. Criminal proceedings have been initiated in many cases where the candidates have tried to cheat/impersonate/ adopt malpractices.
- iii. Strict action has also been taken including CBI Inquiry against the wrong-doers.

2. Over the past few years, NTA has deployed many modern technologies to detect different types of malpractices. NTA has a mechanism to;

- i. Analyze the responses of the treated group with respect to cheating behaviour associated with the exam by observational data, interesting statistical approaches, etc.
- ii. We use a non-intrusive technology to identify collaboration, we analyze the question attempting logs to identify suspicious similarities between submitted responses. NTA also reviews candidate exam logs to identify abnormal answering behaviours and CCTV recordings of such candidates are analyzed to confirm malpractice with evidence.
- iii. Analyze the behaviour of students who passed their answers to other candidates as opposed to candidates who received these answers.
- iv. Use of technologies to reveal cheating behaviour, even after the exams, to reveal the cheating occurrence
- v. Catch the likely cheaters through Artificial Intelligence-based tools.
- vi. We focus on cheating in the form of seeking outside help during the exam and catch hold on this by implying various technologies/tools. The examination centres are continuously being monitored through artificial-enabled systems.

3. It is crucial that you carefully read the following points to avoid facing any action;

- i. Use of electronic devices (including communication tools and accessories) or stationery materials (including bits of paper) not relevant for the said examination is strictly prohibited.
- ii. Candidates should upload their recent and clearly visible images during the application process. Candidates who have uploaded poorquality photographs or not his/her own photograph will be stopped at the gate.
- iii. NTA scrutinizes every candidate trying to appear more than once by submitting multiple application forms.
- iv. Candidates are strictly advised to occupy only the seats allocated to them.
- v. Candidates are advised not to move during the exam including talking/peeping or assisting others in the exam.
- vi. Apart from the biometric attendance and frisking at entry, candidates will be frisked and biometric attendance will be taken again on entry from bio-break/toilets break.
- vii. Documents in relation to examination viz. admit card, rank letter, self-declaration, etc. are subject to verification for any manipulation and fabrication.
- viii. Candidates should not create obstacles in the smooth and fair conduct of the examination.
- ix. Scribe accompanying the Person with Disability (PwD/PwBD) should not solve the questions. The scribe must mark the response chosen by the candidate only. It shall also be ensured that the qualification of the scribe/ Passage Dictator should not be more than the minimum qualification criteria of the examination. However, the qualification of the scribe /Passage Dictator should be matriculate.

"NTA wishes the best for your future, please don't indulge in any malpractice"

Joint Entrance Examination (Main) – 2025 Session 2
Admit Card-Provisional

Declaration from Non-AADHAAR Candidate

To

Director (Exam)
National Testing
Agency Okhla,
Delhi - 110020

Subject – Declaration regarding **JEE (Main) – 2025 Session 2 (April 2025)** Examination- Reg.

Dear Sir/Ma'am,

I hereby declare that I am S/O (Father)

and (Mother)..... declare that my candidature for this examination is genuine and all the details filled in online application form are valid as per my knowledge.

The image uploaded on my application form is not matching with my face due to the following reasons (please tick appropriate option below):

- ☐ Blank Image uploaded with Application form.
☐ Poor quality of image uploaded with Application form.
☐ Incorrect image uploaded with Application form.
☐ Others

Application Id and Roll No

Your Sincerely

Candidate name -

Govt Id proof Type : Govt. ID No :

Date : Shift No. :

Signature of Candidate

Remarks by NTA officials (if candidate found suspicious), tick one of the below options:

- ☐ Candidate does not have an Aadhaar card
☐ Candidate Aadhaar card is not verified

Signature of NTA Observer / Deputy Observer:

Mobile Number of NTA Observer / Deputy Observer:

Copy of Govt. Id proof type taken from candidate:

13-12-2018

Rehearsal Examination 2018 Marking Scheme for class - XII Subject - Physics

Section A:

- 1a) (i) op (A) (ii) op (C) $\rightarrow 3.4 \text{ eV}$ (iii) $eV_0 = h\nu - h\nu_0$ thus,
($470 \pm 5\% \text{ nm}$) as $|E_T| = E_K$ or $V_0 = \frac{h}{e} \nu - \frac{h}{e} \nu_0$
slope $= \frac{h}{e}$; $C = \frac{-h\nu_0}{e}$
 $\therefore \left| \frac{C}{\text{slope}} \right| = \nu_0$ op (D)
- (iv) op (B) opposite phase
(v) op (B) : $\neq 0$ vol.

1b) i) For Ryman Series : $\frac{1}{\lambda} = R_H \left(\frac{1}{1^2} - \frac{1}{n_i^2} \right)$

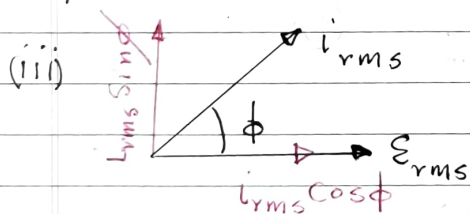
For shortest λ , $n_i = \infty$

or $\frac{1}{\lambda} = 1.097 \times 10^7 \left[1 - \frac{1}{\infty^2} \right]$

$\therefore \lambda = 0.912 \times 10^{-7} \text{ m} = 912 \text{ \AA}$

(vii) $\oint_C \mathbf{B} \cdot d\mathbf{r} = \mu_0 \left[I_c + \epsilon_0 \frac{d\phi_E}{dt} \right]$
 $I_c \rightarrow$ conduction current
 $\phi_E \rightarrow$ Electric flux.

- (ii) This happens when the emf of the driving auxillary battery becomes less than the emf of the cell to be measured.



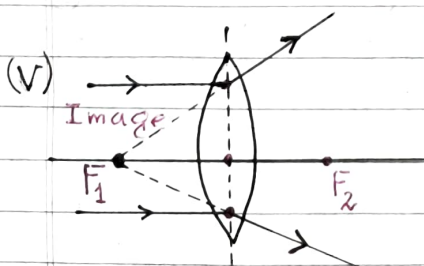
$\therefore$ Power contributed by

$i_{rms} \sin \phi$ is:

$\epsilon_{rms} (i_{rms} \sin \phi) \cos(90^\circ)$

$\therefore i_{rms} \sin \phi$ is Watt-less current.

(iv) $Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{25} \sqrt{\frac{10}{40 \times 10^{-6}}} = \frac{1000}{50} = 20$



vi) Primary Rainbow : Violet inner edge
red colour on the outer edge.

Secondary Rainbow : Red inner edge
violet colour on the outer edge

section B

2) $j = ne v_d$; $\vec{j} = \sigma \vec{E}$

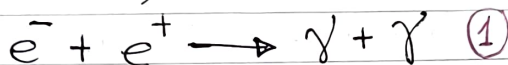
OR

2) $\phi = 5t^3 + t^2 - 5$

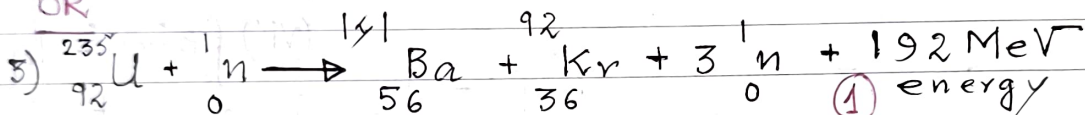
$\epsilon = \frac{-d\phi}{dt} = -15t^2 - 8t$ at $t = 2s$

$\epsilon = -15 \times 2^2 - 8 \times 2 = -76 \text{ volt}^-$

3) ~~When an electron (e^-) and a positron (e^+) collide at low energy pair annihilation occurs followed by the creation of γ -ray photons.~~



OR



Control rods $\rightarrow$ absorb some of the neutrons produced during chain reaction thereby resulting in controlled fission.

4) By ECP:

$$\frac{1}{2} K (r)^2 = \frac{q^2}{4\pi\epsilon} \left[\frac{1}{r} - \frac{1}{2r} \right]$$

or $\frac{K r^2}{2} = \frac{q^2}{4\pi\epsilon} \left(\frac{2-1}{2r} \right)$

or $K = \frac{q^2}{4\pi\epsilon} \times \frac{1}{r^3}$

5) $M = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{2} \right) = -\frac{150}{5} \left(1 + \frac{5}{25} \right) = -33.6$

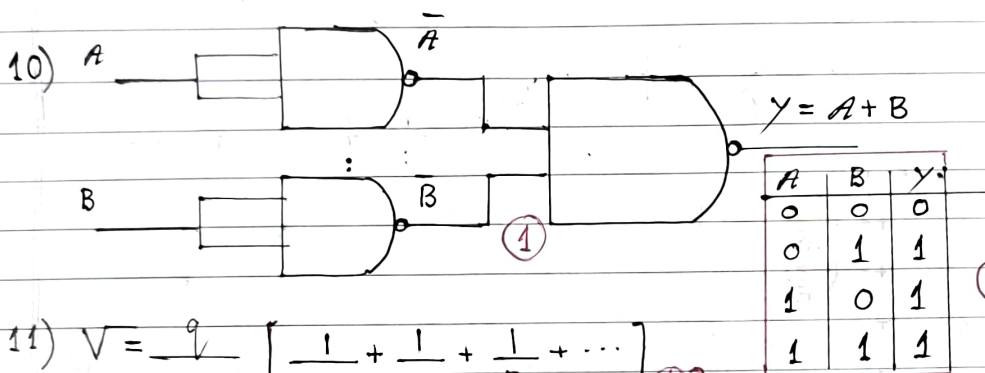
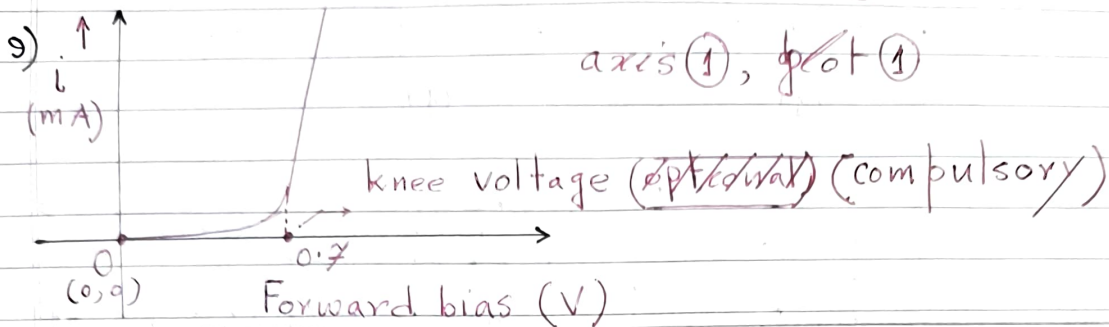
6) 'Interference' is the result of interaction of light waves coming from two different wavefronts originating from 2 coherent sources.

Diffraction is the result of interaction of light waves coming from 2 parts of same wave front.

(1+1)

7) $\beta = \frac{D\lambda}{d}$, $\beta' = \frac{D\lambda}{d\mu}$ ① or $\beta' = \frac{\beta}{\mu} = \frac{2\text{ mm}}{4/3} = 1.5\text{ mm}$ ①

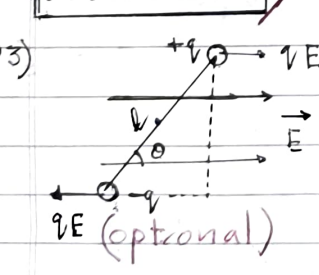
8) $\beta = \left(\frac{\alpha}{1-\alpha}\right) \Rightarrow \alpha = \frac{\beta}{1+\beta}$ or $\alpha = 0.986$ ①



11) $V = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots \right]$ ①
 $E = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$ ①

12) $V_c = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} = \frac{5 \times 10^{-6} \times 16 + 10 \times 10^{-6} \times 13}{15 \times 10^{-6}}$ ①
 $= \frac{210}{15} \text{ V} = 14 \text{ volt}$ ①

Section C

13)  ①

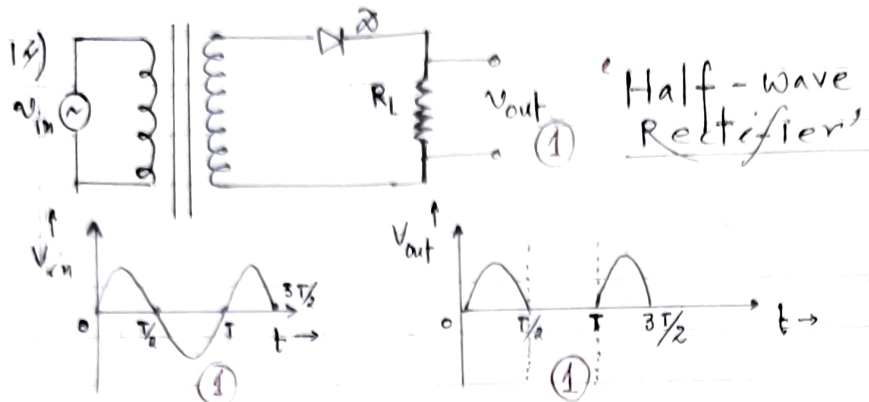
① $\tau = qEl \sin\theta \approx qEl\theta$ (as θ is small)

$\tau_{\text{net}} = I\alpha = \left[m\left(\frac{l}{2}\right)^2 + m\left(\frac{l}{2}\right)^2 \right] \alpha$

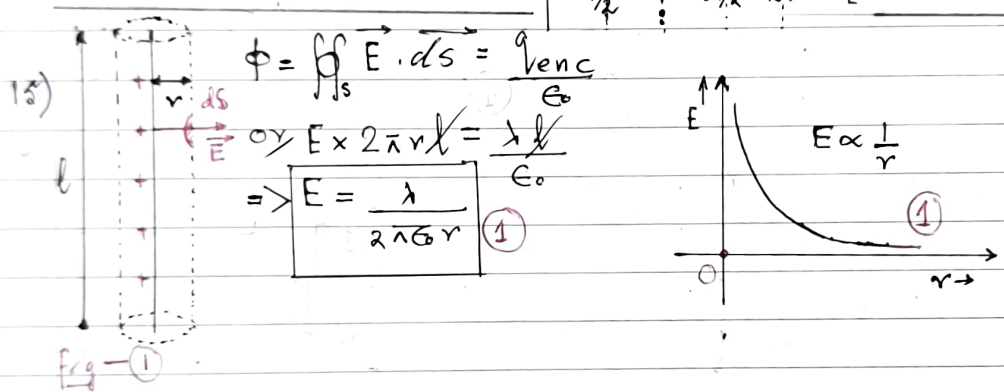
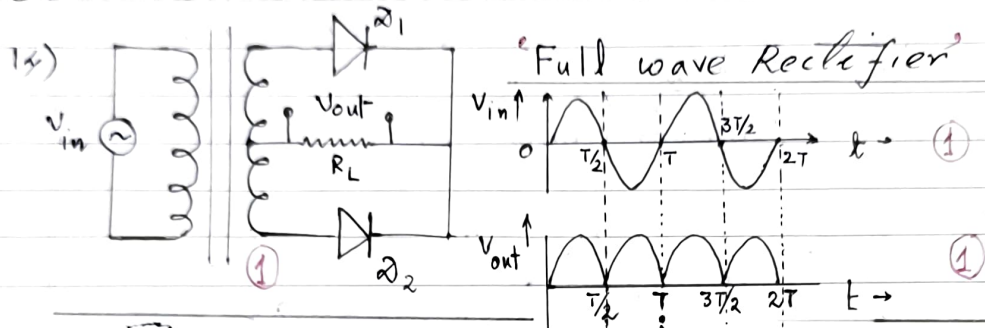
In Equilibrium,

① $\begin{cases} qEl\theta = \frac{ml^2}{2} \frac{d^2\theta}{dt^2} \text{ or } \frac{d^2\theta}{dt^2} = \frac{2qE}{ml} \theta = \omega^2 \theta \\ \therefore \omega = \sqrt{\frac{2qE}{ml}} = \frac{2\pi}{T} \text{ or } T = 2\pi \sqrt{\frac{ml}{2qE}} \end{cases}$

Reqd time = $T = 2\pi \sqrt{\frac{ml}{2qE}}$ ①



OR



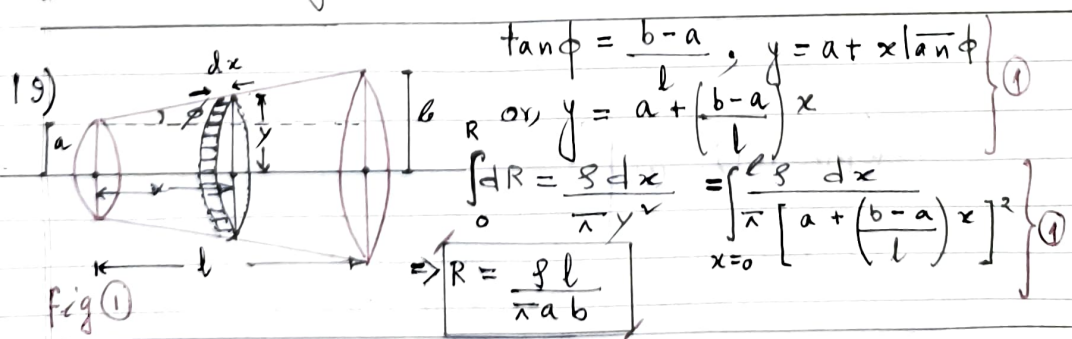
16) Fig ①, Proof ②

17) Fig ②, $\phi = 180^\circ$ or π radian ①

⑱ Proof ②, Significance: $I_{av} \times \frac{T}{2} = \int_0^{T/2} i_0 \sin \omega t dt = q$ ①

OR

⑱ Proof ②, Significance: $I_{rms}^2 R \frac{T}{2} = \int_0^{T/2} i_0^2 \sin^2 \omega t R dt = H$ ①



↳ Integration:

$$\text{Let } a + \frac{(b-a)}{l} x = \bar{x} \quad \text{or, } \frac{(b-a)}{l} dx = d\bar{x}$$

$$\text{When, } x=0, \bar{x}=a \quad \text{and } x=l, \bar{x}=b$$

$$\therefore R = \frac{\rho}{\pi} \frac{l}{(b-a)} \int_a^b \frac{d\bar{x}}{\bar{x}} = \frac{\rho l}{\pi (b-a)} \left[-\frac{1}{\bar{x}} \right]_a^b = \frac{\rho l}{\pi (b-a)} \left(\frac{1}{a} - \frac{1}{b} \right)$$

$$\text{or, } R = \frac{\rho l}{\pi (b-a)} \times \frac{(b-a)}{ab}$$

$$\text{or, } R = \frac{\rho l}{\pi ab}$$

$$20a) E = E_0 \cos(2\pi f t), \quad f = \frac{31.4}{2\pi} = 50 \text{ Hz}$$

$$X_L = 2\pi f L = 2\pi \times 50 \times 0.1 = 31.4 \Omega$$

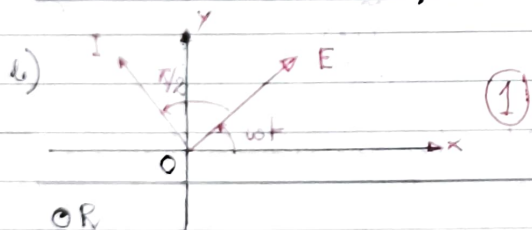
$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 25 \times 10^{-6}} = 127.4 \Omega$$

$$(i) \bar{z} = \sqrt{R^2 + (X_C - X_L)^2} = \sqrt{25^2 + (127.4 - 31.4)^2} = 99.2 \Omega$$

$$(ii) \tan \phi = \frac{X_C - X_L}{R} = \frac{96}{25} = 3.84 \quad \text{or, } \phi = 75.4^\circ$$

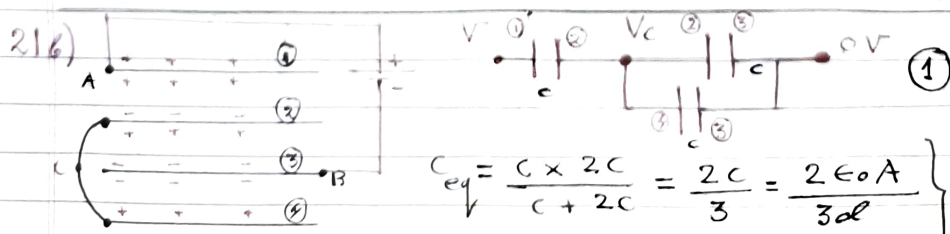
is the phase angle by which i leads E .

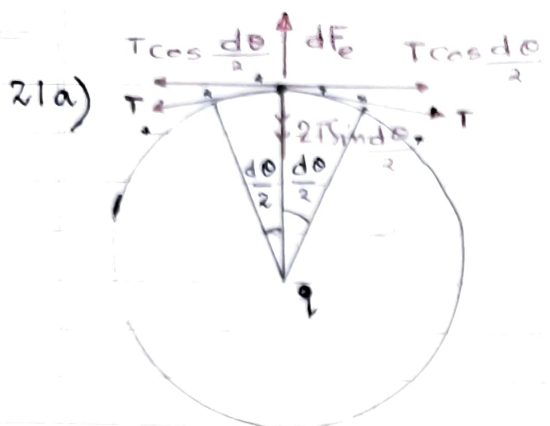
$$(iii) \text{Power factor } \cos \phi = \cos 75.4^\circ = 0.25.$$



$$20a) \text{ Fig (1), Proof (2). } B = \frac{\mu_0 i}{4\pi r} [\sin \phi_1 + \sin \phi_2]$$

$$b) M = \frac{\mu_0 N_1 N_2 A}{l} \quad \text{AT/steps (2)}$$





$$\lambda = \frac{Q}{2\pi R}, \quad dq = \lambda dl = \lambda R d\theta$$

$$\text{or, } dq = \frac{Q}{2\pi} d\theta$$

$$\Rightarrow dq = \frac{Q d\theta}{2\pi} \quad (1)$$

$$\left. \begin{aligned} dF_e &= \frac{1}{4\pi\epsilon_0} \frac{q \cdot Q}{R^2} \times \frac{Q d\theta}{2\pi} \\ &= \frac{Q^2 d\theta}{8\pi^2 \epsilon_0 R^2} \end{aligned} \right\} (1)$$

In Equilibrium: $dF_e = 2T \sin \frac{d\theta}{2} = T d\theta$

$$\text{or, } \frac{Q^2 d\theta}{8\pi^2 \epsilon_0 R^2} = T d\theta$$

$$\text{or, } T = \frac{Q^2}{8\pi^2 \epsilon_0 R^2}$$

(1)

OR

21a) $\frac{-13.6}{n^2} = -3.4 \quad \text{or, } n=2 \quad (1)$

By Bohr's Quantisation condition

$$mvr = \frac{n h}{2\pi} = 2 \times \frac{h}{2\pi} = \frac{h}{\pi}$$

$$\therefore L = \frac{h}{\pi}$$

(1)

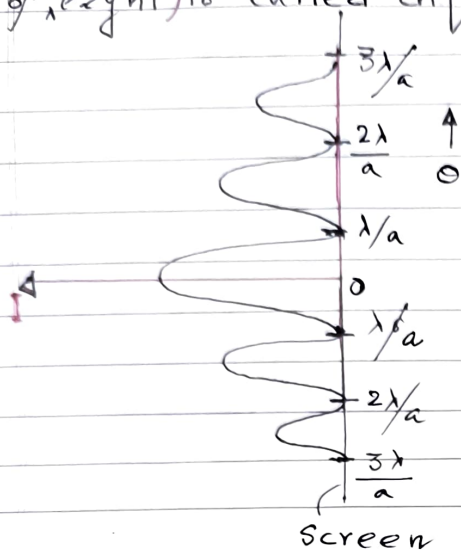
6) $\frac{dN}{dt} = -\lambda N + \alpha \quad (1)$ May be obtained by the given condition.

$$\text{or, } \int_{N_0}^N \frac{dN}{\alpha - \lambda N} = \int_0^t dt \quad \text{or, } \left[\frac{-1}{\lambda} \log_e (\alpha - \lambda N) \right]_{N_0}^N = t$$

$$\text{or, } \frac{1}{\lambda} \log_e \left| \frac{\alpha - \lambda N_0}{\alpha - \lambda N} \right| = t \quad (1)$$

$$\text{or, } N = \left\{ \alpha - (\alpha - \lambda N_0) e^{-\lambda t} \right\} \frac{1}{\lambda} \quad (1)$$

22a) The phenomenon of bending of light round the sharp corners and spreading into the region of geometrical shadow (when the size of the obstacle is comparable to the wavelength of ^{incident} light) is called diffraction of light. (1)



22b) $\mu = \tan \beta$ or, $\beta = \tan^{-1} \mu$ or, $\beta = 60^\circ$ (1)

OR

22a) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ or, $\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = \frac{f+u}{uf}$ or, $v = \frac{uf}{u+f}$ (1)

or, $\frac{dv}{dt} = \frac{(u+f)f \frac{du}{dt} - uf \cdot \frac{du}{dt}}{(u+f)^2}$

(1) or, $\frac{dv}{dt} = \left[\frac{f}{u+f} \right] \frac{du}{dt}$

as, for a convex lens f is +ve, u is (-)ve

$\Rightarrow \frac{dv}{dt} > \frac{du}{dt}$

$\therefore (f+u) < f \Rightarrow \frac{f^2}{(u+f)^2} > 1$ (1)

Thus the image of the ant moves with a greater speed than that of the ant.

22b) $\frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$ (1)

or, $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R} - \left[-\frac{1}{R} \right] \right)$

or, $\frac{1}{f} = (\mu - 1) \times \frac{2}{R}$, ATP, $R = f$

or, $\frac{1}{f} = \frac{2(\mu - 1)}{f}$ or, $\mu - 1 = \frac{1}{2}$ or, $\mu = 1.5$ (1)

